
PH102: Physics-2

Electrodynamics

Lecture # 03
Orthogonal Curvilinear System
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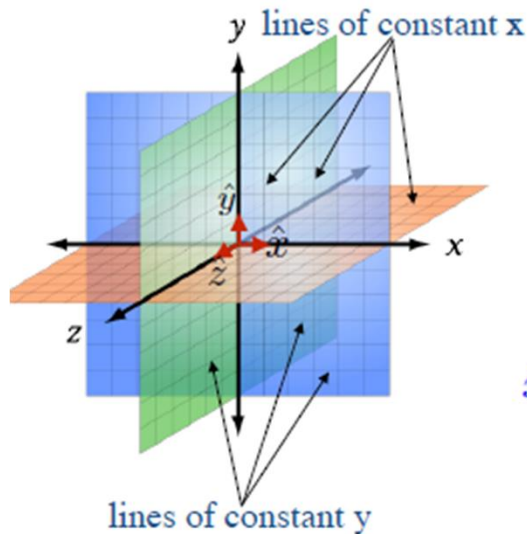
Previous lecture

- ✓ Second derivative of gradient, divergence and curl of a scalar/vector field.
- ✓ Line, surface and volume integrals.
- ✓ Fundamental theorems of gradient, divergence and curl.

Cartesian Coordinate System

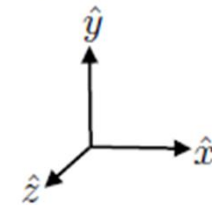
Recall Cartesian: $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$ and $d\vec{r} = dx\hat{x} + dy\hat{y} + dz\hat{z}$, where $\hat{x}, \hat{y}, \hat{z}$ are **constant** unit vectors

$$\left(d\vec{r} = \frac{\partial \vec{r}}{\partial x} dx + \frac{\partial \vec{r}}{\partial y} dy + \frac{\partial \vec{r}}{\partial z} dz \right)$$



Length scales properly match: both LHS and RHS has the dimension of length: $|d\vec{r}| = \sqrt{dx^2 + dy^2 + dz^2}$

Unit vectors in Cartesian Coordinates:



$\hat{x}, \hat{y}, \hat{z}$: constant in direction (direction of increase of x, y and z)
constant in magnitude (norm=1)

Orthogonality: $\hat{x}_i \cdot \hat{x}_j = \delta_{ij}$

Remember also: $\hat{x}_i \times \hat{x}_j = \epsilon_{ijk} \hat{x}_k$.

Suppose we want to go to curvilinear coordinates from Cartesian: $(x, y, z) \rightarrow (u_1, u_2, u_3)$

$$\begin{aligned} \vec{r} &\neq u_1\hat{u}_1 + u_2\hat{u}_2 + u_3\hat{u}_3 \\ d\vec{r} &\neq du_1\hat{u}_1 + du_2\hat{u}_2 + du_3\hat{u}_3 \\ |d\vec{r}| &\neq \sqrt{du_1^2 + du_2^2 + du_3^2} \end{aligned}$$

Think about $u_1 = r, u_2 = \theta, u_3 = \phi$, then the LHS has dimension of length, but RHS does not have the proper dimension.

Curvilinear Coordinate System

In general, a point may be represented by the intersection of three surfaces defined by $u_1 = \text{constant}$, $u_2 = \text{constant}$, and $u_3 = \text{constant}$.

These surfaces may not be planes; they may be curved surfaces.

These surfaces need not be perpendicular to each.

If the point of intersection P is $\vec{l} = x\hat{x} + y\hat{y} + z\hat{z}$

and we write $\vec{l} = \vec{l}(u_1, u_2, u_3)$

or, $x = x(u_1, u_2, u_3)$, $y = y(u_1, u_2, u_3)$, $z = z(u_1, u_2, u_3)$,

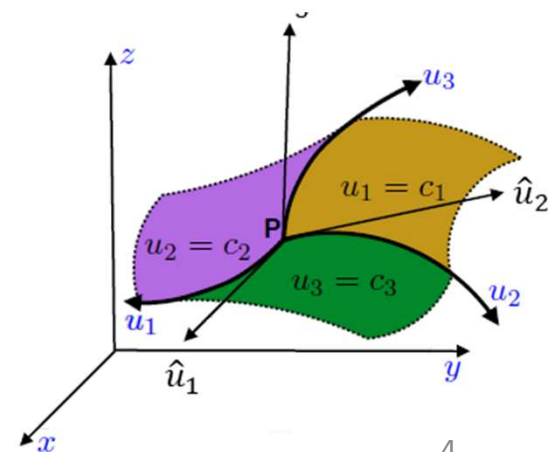
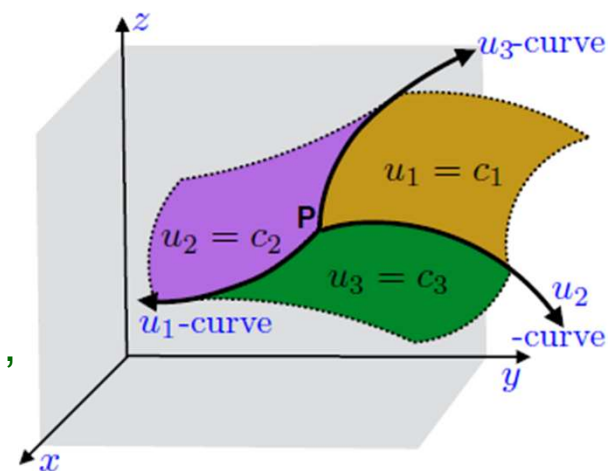
then we can associate a set of coordinates such that

$$u_1 = u_1(x, y, z), u_2 = u_2(x, y, z), u_3 = u_3(x, y, z).$$

Now, from the figure, a tangent vector to the curve u_1 at P

is $\frac{\partial \vec{l}}{\partial u_1}$ and a unit tangent vector along u_1 is

$$\hat{u}_1 = \frac{\frac{\partial \vec{l}}{\partial u_1}}{\left| \frac{\partial \vec{l}}{\partial u_1} \right|} \Rightarrow \frac{\partial \vec{l}}{\partial u_1} = h_1 \hat{u}_1$$



(Contd ...)

Similarly, $\frac{\partial \vec{l}}{\partial u_2} = h_2 \hat{u}_2$ and $\frac{\partial \vec{l}}{\partial u_3} = h_3 \hat{u}_3$

where h_1 , h_2 , and h_3 are scale factors. These scale factors are needed as they ensure that dx , dy , and dz have equivalent derivatives having the unit of length.

We can also write, $d\vec{l} = \frac{\partial \vec{l}}{\partial u_1} du_1 + \frac{\partial \vec{l}}{\partial u_2} du_2 + \frac{\partial \vec{l}}{\partial u_3} du_3$

or, $d\vec{l} = \hat{u}_1 h_1 du_1 + \hat{u}_2 h_2 du_2 + \hat{u}_3 h_3 du_3 = \hat{u}_1 dl_1 + \hat{u}_2 dl_2 + \hat{u}_3 dl_3$

Now, $dl^2 = d\vec{l} \cdot d\vec{l} = \sum_{i,j} \frac{\partial \vec{l}}{\partial u_i} \cdot \frac{\partial \vec{l}}{\partial u_j} du_i du_j = \sum_{i,j} g_{ij} du_i du_j$

where, $g_{ij} = \frac{\partial \vec{l}}{\partial u_i} \cdot \frac{\partial \vec{l}}{\partial u_j}$

For orthogonal curvilinear coordinates, $g_{ij} = 0$ for $i \neq j$.

Therefore,

$$dl^2 = (h_1 du_1)^2 + (h_2 du_2)^2 + (h_3 du_3)^2 = \sum_i (h_i du_i)^2$$

(Contd ...)

In order to arrive at the orthogonal curvilinear coordinates, we imposed the following condition on the basis vectors $\hat{u}_1, \hat{u}_2$ and $\hat{u}_3$

$$\begin{aligned}\hat{u}_1 \cdot \hat{u}_2 &= 0, & \hat{u}_2 \cdot \hat{u}_3 &= 0, & \hat{u}_3 \cdot \hat{u}_1 &= 0 \\ \hat{u}_1 \cdot \hat{u}_1 &= 1, & \hat{u}_2 \cdot \hat{u}_2 &= 1, & \hat{u}_3 \cdot \hat{u}_3 &= 1\end{aligned}$$

They are also arranged in such a way that the following relations are satisfied:

$$\hat{u}_1 \times \hat{u}_2 = \hat{u}_3, \quad \hat{u}_2 \times \hat{u}_3 = \hat{u}_1, \quad \hat{u}_3 \times \hat{u}_1 = \hat{u}_2$$

The surface area element $d\mathbf{a}$ normal to $\hat{u}_1, \hat{u}_2$ and $\hat{u}_3$ can be written as follows.

$$\begin{aligned}d\mathbf{a}_1 &= \hat{u}_1 da_1 = d\mathbf{l}_2 \times d\mathbf{l}_3 = (h_2 h_3 du_2 du_3) \hat{u}_1 \\ d\mathbf{a}_2 &= \hat{u}_2 da_2 = d\mathbf{l}_3 \times d\mathbf{l}_1 = (h_3 h_1 du_3 du_1) \hat{u}_2 \\ d\mathbf{a}_3 &= \hat{u}_3 da_3 = d\mathbf{l}_1 \times d\mathbf{l}_2 = (h_1 h_2 du_1 du_2) \hat{u}_3\end{aligned}$$

The corresponding magnitudes of surface area elements are respectively,

$$da_1 = h_2 h_3 du_2 du_3 \quad da_2 = h_3 h_1 du_3 du_1 \quad da_3 = h_1 h_2 du_1 du_2$$

The differential volume element can be written as $d\tau = h_1 h_2 h_3 du_1 du_2 du_3$

Differential Vector Operations

Let us express the gradient, divergence and curl of a vector field in terms of the coordinates u_1, u_2 and u_3 . These coordinates are orthogonal in nature.

Gradient:

The gradient vector in terms of infinitesimal displacements dl_1, dl_2, dl_3 along the three orthogonal directions $\hat{u}_1, \hat{u}_2, \hat{u}_3$ in the generalized coordinates, u_1, u_2 and u_3 can be written as,

$$\nabla = \hat{u}_1 \frac{\partial}{\partial l_1} + \hat{u}_2 \frac{\partial}{\partial l_2} + \hat{u}_3 \frac{\partial}{\partial l_3}$$

The gradient of a scalar, $T(u_1, u_2, u_3)$ is written as,

$$\begin{aligned} \nabla T &= \left(\hat{u}_1 \frac{\partial}{\partial l_1} + \hat{u}_2 \frac{\partial}{\partial l_2} + \hat{u}_3 \frac{\partial}{\partial l_3} \right) T = \hat{u}_1 \frac{\partial T}{\partial l_1} + \hat{u}_2 \frac{\partial T}{\partial l_2} + \hat{u}_3 \frac{\partial T}{\partial l_3} \\ &= \hat{u}_1 \frac{1}{h_1} \frac{\partial T}{\partial u_1} + \hat{u}_2 \frac{1}{h_2} \frac{\partial T}{\partial u_2} + \frac{1}{h_3} \hat{u}_3 \frac{\partial T}{\partial u_3} = \sum_i \hat{u}_i \frac{1}{h_i} \frac{\partial T}{\partial u_i} \end{aligned}$$

The variation of the function dT is

$$dT = \nabla T \cdot d\mathbf{l} = \sum_i \hat{u}_i \frac{1}{h_i} \frac{\partial T}{\partial u_i} \cdot dl_i \hat{u}_i = \sum_i \hat{u}_i \frac{1}{h_i} \frac{\partial T}{\partial u_i} \cdot h_i du_i \hat{u}_i = \sum_i \frac{\partial T}{\partial u_i} du_i$$

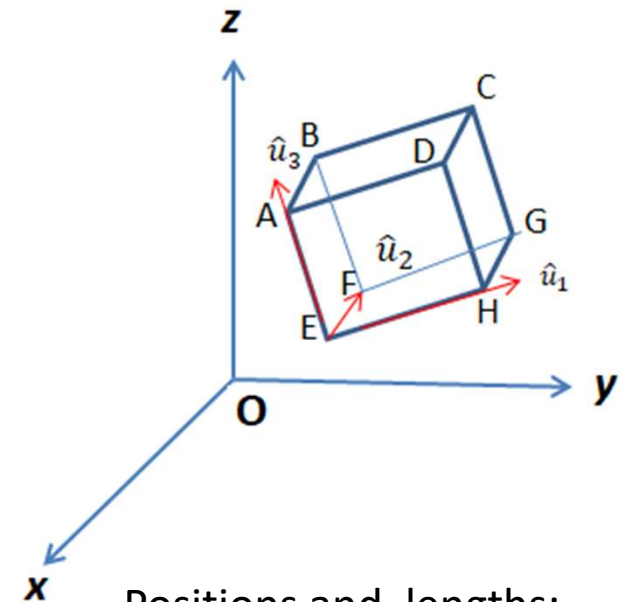
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Divergence: The divergence operator in generalized orthogonal curvilinear coordinate (OCC) can be defined by using Gauss theorem as follows,

$$\nabla \cdot \mathbf{v} = \lim_{d\tau \rightarrow 0} \frac{\oint \mathbf{v} \cdot d\mathbf{a}}{d\tau} \quad (1)$$

So, divergence at a point can be obtained by considering an infinitesimal volume $d\tau$ around the point and calculating the total flux leaving the closed surface which bound the volume.

Consider a small cubical volume in OCC system as shown. The total flux leaving the volume can be determined by considering the net flux leaving the opposite surfaces and adding them all. (surfaces are shown as plane but they can be curved as well)



Positions and lengths:

E : u_1, u_2, u_3

H : $(u_1 + du_1), u_2, u_3$

F : $u_1, (u_2 + du_2), u_3$

A : $u_1, u_2, (u_3 + du_3)$

$EH = dl_1 = h_1 du_1$

$EF = dl_2 = h_2 du_2$

$EA = dl_3 = h_3 du_3$

(Contd ...)

The vector function $\mathbf{v}$ can be taken as,

$$\mathbf{v} = v_1 \hat{u}_1 + v_2 \hat{u}_2 + v_3 \hat{u}_3 \quad (2)$$

The elemental area of the surface, ABFE, where u_1 is constant, is

$$(d\mathbf{a})_{u_1} = dl_2 dl_3 (-\hat{u}_1) = (h_2 du_2)(h_3 du_3)(-\hat{u}_1)$$

The flux through the surface ABFE is, $\mathbf{v} \cdot d\mathbf{a} = -(v_1 h_2 h_3) du_2 du_3$ (3)

For the surface DCGH, the elemental area is

$$(d\mathbf{a})_{u_1+du_1} = dl_2 dl_3 (\hat{u}_1) = (h_2 du_2)(h_3 du_3)(\hat{u}_1)$$

The flux through the surface DCGH is, $\mathbf{v} \cdot d\mathbf{a} = (v_1 h_2 h_3)_{u_1+du_1} du_2 du_3$ (4)

(u_1+du_1 is constant for DCGH surface, so it is shown as the suffix)

The net flux coming out of surfaces ABFE and DCGH is obtained by adding (3) and (4),

$$\mathbf{v} \cdot d\mathbf{a} = (v_1 h_2 h_3)_{u_1+du_1} du_2 du_3 - (v_1 h_2 h_3)_{u_1} du_2 du_3 = \left\{ \left[\frac{\partial}{\partial u_1} (v_1 h_2 h_3) \right] du_1 \right\} du_2 du_3$$

where we have used the fact that $f(x + dx) - f(x) = \left(\frac{\partial f}{\partial x} \right) dx$

(Contd ...)

We know that $d\tau = dl_1 dl_2 dl_3 = h_1 h_2 h_3 du_1 du_2 du_3$ (5)

By using the above equality, the net flux through the pair of surfaces ABFE and DCGH is

$$\mathbf{v} \cdot d\mathbf{a} = \left[\frac{\partial}{\partial u_1} (v_1 h_2 h_3) \right] \frac{d\tau}{h_1 h_2 h_3} \quad (6)$$

Similarly the net flux coming out of surfaces ADHE and BCGF is

$$\mathbf{v} \cdot d\mathbf{a} = \left[\frac{\partial}{\partial u_2} (v_2 h_3 h_1) \right] \frac{d\tau}{h_1 h_2 h_3} \quad (7)$$

and for surfaces FGHE and BCDA is $\mathbf{v} \cdot d\mathbf{a} = \left[\frac{\partial}{\partial u_3} (v_3 h_1 h_2) \right] \frac{d\tau}{h_1 h_2 h_3}$ (8)

Adding eqs. (6) to (8), the net flux coming out of the closed surface is obtained,

$$\oint \mathbf{v} \cdot d\mathbf{a} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} (v_1 h_2 h_3) + \frac{\partial}{\partial u_2} (v_2 h_3 h_1) + \frac{\partial}{\partial u_3} (v_3 h_1 h_2) \right] d\tau \quad (9)$$

Substitution of eq.(9) in the definition of divergence, eq.(1), we get

$$\nabla \cdot \mathbf{v} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} (v_1 h_2 h_3) + \frac{\partial}{\partial u_2} (v_2 h_3 h_1) + \frac{\partial}{\partial u_3} (v_3 h_1 h_2) \right]$$

(Contd ...)

Curl: The Curl operator in OCC can be defined by using Strokes theorem. They are given in component form as follows,

$$(\nabla \times \mathbf{v})_{u_1} = \lim_{da \rightarrow 0} \frac{\oint_{u_1} \mathbf{v} \cdot d\mathbf{l}}{da} \quad (10)$$

Here, the path is chosen such that the infinitesimal surface area enclosed by the path is normal to $\hat{u}_1$. Similar expression can be written for other two components. Let us consider a small surface whose normal is along $\hat{u}_1$ as shown in the figure.

Positions:

A: u_1, u_2, u_3

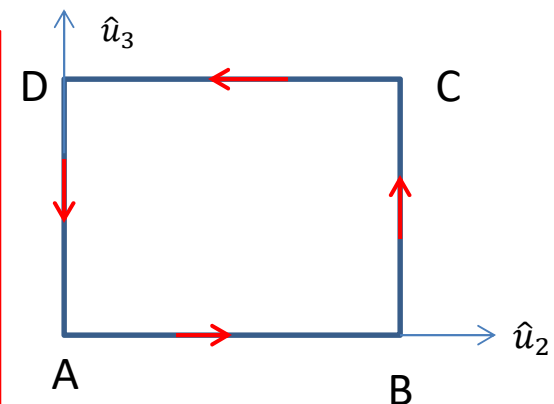
Path AB: $d\mathbf{l} = dl_2 \hat{u}_2 = h_2 du_2 \hat{u}_2$

B: $u_1, (u_2 + du_2), u_3$

Path CD: $d\mathbf{l} = dl_2(-\hat{u}_2) = -h_2 du_2 \hat{u}_2$

C: $u_1, (u_2 + du_2), (u_3 + du_3)$; Path BC: $d\mathbf{l} = dl_3(\hat{u}_3) = h_3 du_3 \hat{u}_3$

D: $u_1, u_2, (u_3 + du_3)$; Path DA: $d\mathbf{l} = dl_3(-\hat{u}_3) = -h_3 du_3 \hat{u}_3$



(Contd ...)

For path AB, $\mathbf{v} \cdot d\mathbf{l} = (v_2 h_2)_{u_3} du_2$ and CD, $\mathbf{v} \cdot d\mathbf{l} = -(v_2 h_2)_{u_3+du_3} du_2$

So for the paths AB and CD,

$$\mathbf{v} \cdot d\mathbf{l} = -(v_2 h_2)_{u_3+du_3} du_2 + (v_2 h_2)_{u_3} du_2 = - \left[\frac{\partial(v_2 h_2)}{\partial u_3} \right] du_3 du_2 \quad (11)$$

Similarly for the path BC and DA,

$$\mathbf{v} \cdot d\mathbf{l} = (v_3 h_3)_{u_2+du_2} du_3 \quad \text{and} \quad \mathbf{v} \cdot d\mathbf{l} = -(v_3 h_3)_{u_2} du_3$$

So, for the paths BC and DA,

$$\mathbf{v} \cdot d\mathbf{l} = (v_3 h_3)_{u_2+du_2} du_3 - (v_3 h_3)_{u_2} du_3 = \left[\frac{\partial(v_3 h_3)}{\partial u_2} \right] du_2 du_3 \quad (12)$$

Therefore, for the closed path ABCD, the line integral is obtained by adding eqs.

(11) and (12).

$$\oint \mathbf{v} \cdot d\mathbf{l} = \left[\frac{\partial(h_3 v_3)}{\partial u_2} \right] du_2 du_3 - \left[\frac{\partial(h_2 v_2)}{\partial u_3} \right] du_2 du_3 = \left[\frac{\partial(h_3 v_3)}{\partial u_2} - \frac{\partial(h_2 v_2)}{\partial u_3} \right] du_2 du_3 \quad (13)$$

(Contd ...)

We know that, $da = (h_2 h_3) du_2 du_3$

$$\oint \mathbf{v} \cdot d\mathbf{l} = \frac{1}{h_2 h_3} \left[\frac{\partial(h_3 v_3)}{\partial u_2} - \frac{\partial(h_2 v_2)}{\partial u_3} \right] da \quad (14)$$

Substituting (26) in the definition of Curl V, (eq.22)

$$(\nabla \times \mathbf{v})_{u_1} = \frac{1}{h_2 h_3} \left[\frac{\partial(h_3 v_3)}{\partial u_2} - \frac{\partial(h_2 v_2)}{\partial u_3} \right] \quad (15)$$

By choosing surfaces normal to other two axes, the other components of Curl can be obtained and they are given as follows,

$$(\nabla \times \mathbf{v})_{u_2} = \frac{1}{h_3 h_1} \left[\frac{\partial(h_1 v_1)}{\partial u_3} - \frac{\partial(h_3 v_3)}{\partial u_1} \right] \quad (16) \quad \text{and} \quad (\nabla \times \mathbf{v})_{u_3} = \frac{1}{h_1 h_2} \left[\frac{\partial(h_2 v_2)}{\partial u_1} - \frac{\partial(h_1 v_1)}{\partial u_2} \right] \quad (17)$$

By combining three components, the Curl can be shown in determinant form

$$\nabla \times \mathbf{v} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} \hat{u}_1 h_1 & \hat{u}_2 h_2 & \hat{u}_3 h_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1 v_1 & h_2 v_2 & h_3 v_3 \end{vmatrix}$$

SUMMARY

$$✓ \quad d\vec{l} = \hat{u}_1 dl_1 + \hat{u}_2 dl_2 + \hat{u}_3 dl_3 \quad \text{where} \quad dl_i = h_i du_i$$

$$✓ \quad d\mathbf{a}_1 = \hat{u}_1 da_1 = d\mathbf{l}_2 \times d\mathbf{l}_3 = (h_2 h_3 du_2 du_3) \hat{u}_1$$

$$d\mathbf{a}_2 = \hat{u}_2 da_2 = d\mathbf{l}_3 \times d\mathbf{l}_1 = (h_3 h_1 du_3 du_1) \hat{u}_2$$

$$d\mathbf{a}_3 = \hat{u}_3 da_3 = d\mathbf{l}_1 \times d\mathbf{l}_2 = (h_1 h_2 du_1 du_2) \hat{u}_3$$

$$✓ \quad d\tau = h_1 h_2 h_3 du_1 du_2 du_3$$

$$✓ \quad \nabla T = \hat{u}_1 \frac{\partial T}{\partial l_1} + \hat{u}_2 \frac{\partial T}{\partial l_2} + \hat{u}_3 \frac{\partial T}{\partial l_3} = \hat{u}_1 \frac{1}{h_1} \frac{\partial T}{\partial u_1} + \hat{u}_2 \frac{1}{h_2} \frac{\partial T}{\partial u_2} + \frac{1}{h_3} \hat{u}_3 \frac{\partial T}{\partial u_3}$$

$$✓ \quad \nabla \cdot \mathbf{v} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} (v_1 h_2 h_3) + \frac{\partial}{\partial u_2} (v_2 h_3 h_1) + \frac{\partial}{\partial u_3} (v_3 h_1 h_2) \right]$$

$$✓ \quad \nabla \times \mathbf{v} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} \hat{u}_1 h_1 & \hat{u}_2 h_2 & \hat{u}_3 h_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1 v_1 & h_2 v_2 & h_3 v_3 \end{vmatrix}$$

Questions?